Notes on an Alternative Integration Method for Adrian Monninger's Model
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Motivation: This memo provides a semi-formal description of a method for computing the end-of-period (marginal) value function in the dynamic microeconomic model used in Adrian Monninger's dissertation (which in turn has a macroeconomic component that isn't touched here). In the model, the agent makes decisions about consumption, saving, and (dis)investment in a durable good; a key complication is that there is a *fixed cost* of adjustment, so there is a "region of inaction" in which he does not adjust his durable quantity. Consequently, the policy function is *discontinuous* and the value function is *non-smooth*– the marginal value function is discontinuous at the boundaries of the region of inaction. Computation of the *expected* marginal value (from the perspective of the end of the prior period, before shocks are resolved) using standard numeric integration methods can lead to "phantom" secondary non-concavities in the value function. Here I present a method that should resolve this problem while still being computationally tractable.

Setup: Rather than specify an entire model, the details of which I will surely get wrong, I will focus only on the interperiod dynamics. Suppose the problem has been solved back to period t, and the value function over decision-time states (m_t, n_t) has been constructed as $\mathrm{v}_t(\cdot)$, where m_t is (normalized) market resources and n_t is (normalized) pre-investment durables. The econometrician now wants to compute the end-of-period (marginal) value function over (normalized) assets a_{t-1} and (normalized) post-investment durables q_{t-1}. All normalizations are with respect to permanent income. The intertemporal dynamics are:

$$\begin{aligned} b_t &= (\mathsf{R}/G_t)a_{t-1}, \qquad G_t \equiv \Gamma\psi_t, \\ m_t &= b_t + \theta_t, \qquad n_t = ((1-\delta)/G_t)q_{t-1}, \\ \psi_t &\sim F_{\psi t}, \qquad \theta_t \sim F_{\theta t}, \qquad \mathbb{E}[\psi_t] = \mathbb{E}[\theta_t] = 1. \end{aligned} \tag{1}$$

That is, permanent income is expected to grow by factor Γ, but is subject to a mean one shock ψ_t, while transitory income shocks θ_t are also mean one. Assets accumulate interest at risk free factor R, while post-decision durables depreciate at rate δ. The econometrician wants to construct an accurate representation of the end-of-period (marginal) value function:

$$\begin{aligned} \mathfrak{v}_{t-1}(a_{t-1}, q_{t-1}) &= \beta\mathbb{E}_{\psi\theta}[G_t^{1-\rho}\mathrm{v}_t(m_t, n_t) \mid a_{t-1}, q_{t-1}], \\ \mathfrak{v}^a_{t-1}(a_{t-1}, q_{t-1}) &= \beta\mathsf{R}\mathbb{E}_{\psi\theta}[G_t^{\rho}\mathrm{v}^m_t(m_t, n_t) \mid a_{t-1}, q_{t-1}], \\ \mathfrak{v}^q_{t-1}(a_{t-1}, q_{t-1}) &= \beta(1-\delta)\mathbb{E}_{\psi\theta}[G_t^{\rho}\mathrm{v}^n_t(m_t, n_t) \mid a_{t-1}, q_{t-1}]. \end{aligned} \tag{2}$$

Method: Define an *intermediate value function* for the split second in time "between" the permanent shock ψ_t resolving and the transitory shock θ_t resolving; assume depreciation of the durable good occurs with the permanent shock.

$$\grave{\mathrm{v}}_t(b_t, n_t) = \mathbb{E}_\theta[\mathrm{v}_t(m_t, n_t) \mid b_t, n_t], \tag{3}$$

$$\mathfrak{v}_{t-1}(a_{t-1}, q_{t-1}) = \beta \mathbb{E}_\psi[G_t^{1-\rho} \grave{\mathrm{v}}_t(b_t, n_t) \mid a_{t-1}, q_{t-1}].$$

Note how the subscript on the expectations operator changes to indicate which shocks are being integrated out in each expression. Here, we simply split the interperiod dynamics into two steps rather than one.

In a standard numeric integral, we would choose some discretized approximation (quadrature) of $F_{\theta t}$ as $\{(\theta_k, p_k)\}_{k=1}^K$ and compute the following sum:

$$\mathbb{E}_\theta[\mathrm{v}_t(m_t, n_t) \mid b_t, n_t] = \int \mathrm{v}_t(b_t + \theta, n_t) dF_{\theta t}(\theta) \approx \sum_{k=1}^K \mathrm{v}_t(b_t + \theta_k, n_t) p_k. \tag{4}$$

However, this is the exact procedure we want to *avoid*, as marginal value is discontinuous.

Instead, consider an alternate approximation to the expectation over θ_t, which fixes a grid of m_t values $\{m_j\}_{j=1}^J$ and lets the *probability* of each $b_t \to m_j$ transition to vary with b_t:

$$\grave{\mathrm{v}}_t(b_t, n_t) = \mathbb{E}_\theta[\mathrm{v}_t(m_t, n_t) \mid b_t, n_t] \approx \sum_{j=1}^J \mathrm{v}_t(m_j, n_t) f_{\theta t}(\underbrace{m_j - b_t}_{=\theta_t}) \Big/ \sum_{j=1}^J f_{\theta t}(m_j - b_t). \tag{5}$$

Here, $f_{\theta t}(\cdot)$ is the pdf associated with the transitory shock cdf $F_{\theta t}(\cdot)$– its derivative with respect to θ_t. Rather than fix a particular discretization of θ_t values with fixed probabilities, we instead compute new probabilities for each $b_t \to m_j$ transition.

This fact imposes an important restriction on the method: the pdf *must actually exist*, and it doesn't in the important case where there are point masses in $F_{\theta t}$, e.g. for unemployment. While it is tempting to set aside such point masses and incorporate them into the expectation in a substep (after computing the expectation of the continuous part of the distribution), the entire premise is that we want to *avoid* lumpy future realizations when computing expectations. If the true distribution *is actually lumpy*, the issue is much worse.

Once the intermediate (marginal) value function $\grave{\mathrm{v}}_t(b_t, n_t)$ has been constructed, computation for constructing the end-of-period (marginal) value function $\mathfrak{v}_{t-1}(a_{t-1}, q_{t-1})$ can use the typical approach of a discretized approximation to $F_{\psi t}$, given as $\{(\psi_k, p_k)\}_{k=1}^K$.

$$\mathfrak{v}_{t-1}(a_{t-1}, q_{t-1}) \approx \beta \sum_{k=1}^K (\Gamma \psi_k)^{1-\rho} \grave{\mathrm{v}}_t \left(\frac{\mathsf{R}}{\Gamma \psi_k} a_{t-1}, \frac{1-\delta}{\Gamma \psi_k} q_{t-1} \right) p_k. \tag{6}$$

Algorithm: To put this method into action, you should do the following:

1. In a pre-solution step, specify post-decision state grids for a and q, as normal.
2. In a pre-solution step, specify fairly dense grids of m and b, denser than the grid of a.
3. In that pre-solution step, for each $F_{\theta t}$ (i.e. just once if this is an infinite horizon problem), calculate $f_{\theta t}(m - b)$ on the cross product of the b-grid and m-grid. Put them in a matrix that has b values by row and m values by column.
4. Still in the pre-solution step, for each b-row in the matrix, take the row-wise sum and divide the row by it. This represents a Markov transition matrix from the exogenous b-grid to the exogenous m-grid.
5. To compute expectations during backwards solution, evaluate $v_t(m_t, n_t)$ on the cross product of the dense m-grid and some grid of n. The grid of n need not be any denser than the one on which you solved the problem, and can be the very same one. Put these (marginal) values into a matrix that has the same n_t value within a column.
6. Pre-multiply the Markov matrix by the matrix you just constructed. The resulting matrix of intermediate expected values will have b row-wise and n column-wise.
7. Use ordinary methods to construct a representation of the intermediate (marginal) value function on the exogenous grids of b and n.
8. Using the pre-specified exogenous grids of a and q, compute expected end-of-period (marginal) value using the typical discretized approximation to the permanent shock distribution. For any (a, q) end-of-period state, there is a finite set of future (b, n) points that can be reached under the discretization.
9. Proceed normally from here with the EGM or quasi-EGM approach you're using.

Conclusion: For brevity (read: out of laziness), I only provided marginal value expressions once, using the value function as the example math throughout. The same numeric integration method can be used for both marginal value functions, but you must be careful to apply the correct coefficients in each step to account for the permanent income normalization. If you have difficulty with this step, I can expand the math above to make it more explicit. Let me know if you have further questions or need clarification, but I think this should be helpful to you in any case.